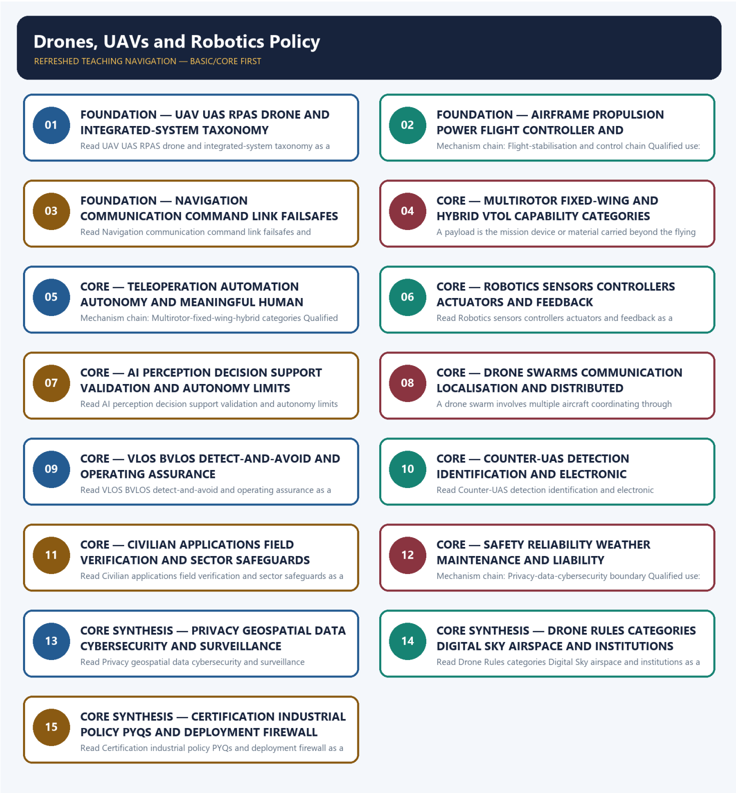
SOLVED PRACTICE WORKBOOK

Drones, UAVs and Robotics Policy - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies UAV-UAS-RPAS-drone boundary?

A. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. B. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. C. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. D. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone.

Answer: A. Explanation: A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of UAV-UAS-RPAS-drone boundary?

A. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. B. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. C. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. D. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery.

Answer: B. Explanation: A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses UAV-UAS-RPAS-drone boundary without changing its institution, unit or status?

A. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. B. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. C. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. D. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.

Answer: C. Explanation: A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about UAV-UAS-RPAS-drone boundary?

A. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. B. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. C. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. D. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category.

Answer: D. Explanation: A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Integrated UAS component stack?

A. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. B. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. C. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. D. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.

Answer: A. Explanation: A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Integrated UAS component stack?

A. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. B. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. C. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. D. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity.

Answer: B. Explanation: A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Integrated UAS component stack without changing its institution, unit or status?

A. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. B. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. C. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. D. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity.

Answer: C. Explanation: A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Integrated UAS component stack?

A. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. B. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. C. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. D. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone.

Answer: D. Explanation: A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Flight-stabilisation and control chain?

A. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. B. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. C. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. D. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.

Answer: A. Explanation: Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Flight-stabilisation and control chain?

A. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. B. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. C. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. D. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity.

Answer: B. Explanation: Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Flight-stabilisation and control chain without changing its institution, unit or status?

A. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. B. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. C. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. D. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop.

Answer: C. Explanation: Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path

state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Flight-stabilisation and control chain?

A. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. B. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. C. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. D. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery.

Answer: D. Explanation: Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Navigation-command-link boundary?

A. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. B. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. C. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. D. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability.

Answer: A. Explanation: Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Navigation-command-link boundary?

A. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. B. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. C. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. D. Multirotors use several rotors and suit hover and confined-area work; fixed-wing

UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity.

Answer: B. Explanation: Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Navigation-command-link boundary without changing its institution, unit or status?

A. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. B. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. C. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: C. Explanation: Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Navigation-command-link boundary?

A. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. B. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. C. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. D. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.

Answer: D. Explanation: Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Payload-mission-capability link?

A. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. B. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. C. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. D. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity.

Answer: A. Explanation: A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Payload-mission-capability link?

A. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. B. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. C. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: B. Explanation: A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Payload-mission-capability link without changing its institution, unit or status?

A. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. B. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. C. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: C. Explanation: A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability.

The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Payload-mission-capability link?

A. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. B. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. C. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. D. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability.

Answer: D. Explanation: A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Multirotor-fixed-wing-hybrid categories?

A. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. B. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. C. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. D. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop.

Answer: A. Explanation: Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Multirotor-fixed-wing-hybrid categories?

A. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. B. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. C. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. D. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop.

Answer: B. Explanation: Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Multirotor-fixed-wing-hybrid categories without changing its institution, unit or status?

A. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. B. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. C. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: C. Explanation: Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Multirotor-fixed-wing-hybrid categories?

A. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. B. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. C. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. D. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity.

Answer: D. Explanation: Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Remote operation-automation-autonomy boundary?

A. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. B. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. C. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: A. Explanation: Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Remote operation-automation-autonomy boundary?

A. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. B. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. C. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: B. Explanation: Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Remote operation-automation-autonomy boundary without changing its institution, unit or status?

A. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. B. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. C. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. D. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation.

Answer: C. Explanation: Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Remote operation-automation-autonomy boundary?

A. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. B. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. C. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. D. Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous.

Answer: D. Explanation: Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Robotics sensing-actuation-control loop?

A. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. B. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. C. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: A. Explanation: Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Robotics sensing-actuation-control loop?

A. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. B. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. C. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. D. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure.

Answer: B. Explanation: Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Robotics sensing-actuation-control loop without changing its institution, unit or status?

A. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. B. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. C. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. D. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service.

Answer: C. Explanation: Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Robotics sensing-actuation-control loop?

A. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. B. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. C. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. D. Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop.

Answer: D. Explanation: Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies AI-enabled autonomy boundary?

A. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. B. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. C. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. D. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation.

Answer: A. Explanation: Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of AI-enabled autonomy boundary?

A. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. B. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. C. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. D. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation.

Answer: B. Explanation: Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses AI-enabled autonomy boundary without changing its institution, unit or status?

A. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. B. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. C. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. D. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone.

Answer: C. Explanation: Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about AI-enabled autonomy boundary?

A. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. B. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. C. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. D. Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.

Answer: D. Explanation: Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Drone-swarm coordination?

A. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. B. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. C. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. D. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards.

Answer: A. Explanation: A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Drone-swarm coordination?

A. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. B. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. C. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. D. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation.

Answer: B. Explanation: A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Drone-swarm coordination without changing its institution, unit or status?

A. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. B. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. C. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. D. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone.

Answer: C. Explanation: A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Drone-swarm coordination?

A. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. B. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. C. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. D. A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure.

Answer: D. Explanation: A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies VLOS-BVLOS operating boundary?

A. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. B. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. C. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. D. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards.

Answer: A. Explanation: VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of VLOS-BVLOS operating boundary?

A. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. B. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. C. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. D. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone.

Answer: B. Explanation: VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses VLOS-BVLOS operating boundary without changing its institution, unit or status?

A. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. B. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. C. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. D. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience.

Answer: C. Explanation: VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about VLOS-BVLOS operating boundary?

A. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. B. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. C. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. D. VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service.

Answer: D. Explanation: VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Counter-UAS and electronic-countermeasure layers?

A. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. B. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. C. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. D. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards.

Answer: A. Explanation: Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Counter-UAS and electronic-countermeasure layers?

A. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. B. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. C. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. D. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience.

Answer: B. Explanation: Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Counter-UAS and electronic-countermeasure layers without changing its institution, unit or status?

A. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. B. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. C. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. D. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience.

Answer: C. Explanation: Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Counter-UAS and electronic-countermeasure layers?

A. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. B. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. C. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. D. Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation.

Answer: D. Explanation: Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Civilian application and evidence boundary?

A. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. B. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. C. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. D. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval.

Answer: A. Explanation: Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Civilian application and evidence boundary?

A. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. B. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. C. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. D. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience.

Answer: B. Explanation: Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Civilian application and evidence boundary without changing its institution, unit or status?

A. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. B. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. C. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. D. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval.

Answer: C. Explanation: Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Civilian application and evidence boundary?

A. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. B. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. C. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. D. Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards.

Answer: D. Explanation: Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Safety-reliability-liability chain?

A. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. B. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. C. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. D. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system.

Answer: A. Explanation: Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Safety-reliability-liability chain?

A. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. B. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. C. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. D. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions.

Answer: B. Explanation: Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Safety-reliability-liability chain without changing its institution, unit or status?

A. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. B. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. C. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. D. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions.

Answer: C. Explanation: Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Safety-reliability-liability chain?

A. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. B. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. C. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. D. Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone.

Answer: D. Explanation: Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Privacy-data-cybersecurity boundary?

A. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. B. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. C. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. D. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval.

Answer: A. Explanation: Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Privacy-data-cybersecurity boundary?

A. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. B. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. C. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. D. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions.

Answer: B. Explanation: Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Privacy-data-cybersecurity boundary without changing its institution, unit or status?

A. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. B. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. C. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. D. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment.

Answer: C. Explanation: Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Privacy-data-cybersecurity boundary?

A. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. B. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. C. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. D. Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience.

Answer: D. Explanation: Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Drone Rules category spine?

A. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. B. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. C. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. D. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment.

Answer: A. Explanation: The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Drone Rules category spine?

A. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. B. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. C. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. D. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status.

Answer: B. Explanation: The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Drone Rules category spine without changing its institution, unit or status?

A. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. B. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. C. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. D. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status.

Answer: C. Explanation: The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Drone Rules category spine?

A. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. B. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. C. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. D. The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval.

Answer: D. Explanation: The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Digital Sky-airspace-institution boundary?

A. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. B. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. C. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. D. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status.

Answer: A. Explanation: MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Digital Sky-airspace-institution boundary?

A. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. B. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. C. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. D. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment.

Answer: B. Explanation: MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Digital Sky-airspace-institution boundary without changing its institution, unit or status?

A. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. B. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. C. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. D. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category.

Answer: C. Explanation: MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Digital Sky-airspace-institution boundary?

A. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. B. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. C. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. D. MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system.

Answer: D. Explanation: MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Certification-registration-pilot-status ladder?

A. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. B. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. C. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. D. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment.

Answer: A. Explanation: Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Certification-registration-pilot-status ladder?

A. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. B. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. C. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. D. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category.

Answer: B. Explanation: Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Certification-registration-pilot-status ladder without changing its institution, unit or status?

A. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. B. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. C. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. D. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone.

Answer: C. Explanation: Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Certification-registration-pilot-status ladder?

A. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. B. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. C. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. D. Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions.

Answer: D. Explanation: Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Dual-use and industrial-policy boundary?

A. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. B. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. C. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. D. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status.

Answer: A. Explanation: Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Dual-use and industrial-policy boundary?

A. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. B. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. C. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. D. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery.

Answer: B. Explanation: Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Dual-use and industrial-policy boundary without changing its institution, unit or status?

A. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. B. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. C. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. D. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.

Answer: C. Explanation: Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Dual-use and industrial-policy boundary?

A. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. B. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. C. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. D. Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment.

Answer: D. Explanation: Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Approval-to-deployment firewall?

A. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. B. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. C. A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. D. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone.

Answer: A. Explanation: Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Approval-to-deployment firewall?

A. A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. B. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. C. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. D. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.

Answer: B. Explanation: Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Approval-to-deployment firewall without changing its institution, unit or status?

A. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. B. Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. C. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. D. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.

Answer: C. Explanation: Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Approval-to-deployment firewall?

A. Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. B. A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. C. Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. D. Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status.

Answer: D. Explanation: Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2020 Prelims applications demand, the 2025 Prelims UAV-capabilities demand and the 2026 provisional-key swarm-coordination and electronic-countermeasure demand directly to this owner. Relevant cross-routes include the 2023 GS-III adversarial-UAV border question and the 2025 GS-I AI-drones-GIS planning question. Three representative cards preserve the core and security routes without inventing any objective answer key.

PYQ DEMAND CARD 1 - 2020 Prelims GS-I

Demand: Assess the routed statements on drone applications in agriculture, volcanic observation and wildlife research.

Status: The official objective key is unavailable locally. The card preserves the technical and farmer-economy cross-route and does not assert an option, answer letter or automatic replacement of field verification.

Model solution: Payload-mission-capability link: A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Civilian application and evidence boundary:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Privacy-data-cybersecurity boundary:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 1 - 2020 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Payload-mission-capability link: A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Civilian application and evidence boundary:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Privacy-data-cybersecurity boundary:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the routed statements on drone applications in agriculture, volcanic observation and wildlife research. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: The official objective key is unavailable locally. The card preserves the technical and farmer-economy cross-route and does not assert an option, answer letter or automatic replacement of field verification. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Payload-mission-capability link: A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Civilian application and evidence boundary:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Privacy-data-cybersecurity boundary:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2020 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2025 Prelims GS-I

Demand: Assess the routed capabilities and limitations of different Unmanned Aerial Vehicle categories.

Status: The official Set-A key is available locally but is not reproduced. The route compares platform architecture and mission fit without inventing range, endurance, payload or objective answers.

Model solution: Integrated UAS component stack: A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. **Payload-mission-capability link:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Multirotor-fixed-wing-hybrid categories:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Remote operation-automation-autonomy boundary:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 2 - 2025 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Integrated UAS component stack: A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. **Payload-mission-capability link:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Multirotor-fixed-wing-hybrid categories:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Remote operation-automation-autonomy boundary:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the routed capabilities and limitations of different Unmanned Aerial Vehicle categories. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: The official Set-A key is available locally but is not reproduced. The route compares platform architecture and mission fit without inventing range, endurance, payload or objective answers. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Integrated UAS component stack: A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload;

capability and certification concern the integrated operating system rather than the airframe alone. **Payload-mission-capability link:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Multirotor-fixed-wing-hybrid categories:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Remote operation-automation-autonomy boundary:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2025 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2023 and 2026 GS-III and Prelims GS-I

Demand: Analyse adversarial UAV threats across Indian borders and assess drone-swarm communication, autonomous coordination and electronic-countermeasure distinctions.

Status: This representative card combines the audited 2023 Mains cross-route with the 2026 provisional-key objective route. No provisional answer is inferred; the model separates threat vector, swarm coordination, layered detection and authorised mitigation.

Model solution: Drone-swarm coordination: A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Counter-UAS and electronic-countermeasure layers:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Safety-reliability-liability chain:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Privacy-data-cybersecurity boundary:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Dual-use and industrial-policy boundary:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Approval-to-deployment firewall:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 3 - 2023 and 2026 GS-III and Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Drone-swarm coordination: A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Counter-UAS and electronic-countermeasure layers:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Safety-reliability-liability chain:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Privacy-data-cybersecurity boundary:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Dual-use and industrial-policy boundary:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Approval-to-deployment firewall:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** PYQ DEMAND CARD 3 - 2023 and 2026 GS-III and Prelims GS-I **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Demand: Analyse adversarial UAV threats across Indian borders and assess drone-swarm communication, autonomous coordination and electronic-countermeasure distinctions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Drone-swarm coordination: A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Counter-UAS and electronic-countermeasure layers:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Safety-reliability-liability chain:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Privacy-data-cybersecurity boundary:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Dual-use and industrial-policy boundary:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Approval-to-deployment firewall:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 3 - 2023 and 2026 GS-III and Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish UAV, UAS, RPAS and drone, and explain why unmanned does not mean autonomous. Answer in about 150 words.

Model thesis: Claim: UAV-UAS-RPAS-drone boundary. **Named evidence/example:** A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated UAS component stack. **Named evidence/example:** A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category.
- A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone.
- Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous.

Qualified conclusion: Claim: UAV-UAS-RPAS-drone boundary. **Named evidence/example:** A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated UAS component stack. **Named evidence/example:** A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Distinguish UAV, UAS, RPAS and drone, and explain why unmanned does not mean autonomous.…", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: UAV-UAS-RPAS-drone boundary. **Named evidence/example:** A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated UAS component stack. **Named evidence/example:** A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: UAV-UAS-RPAS-drone boundary. **Named evidence/example:** A UAV is the aircraft itself; a UAS combines the aircraft, remote pilot station, command-and-control links and associated elements; an RPAS is a UAS with a remote pilot in command and is therefore explicitly non-autonomous, while 'drone' is a colloquial umbrella rather than a precise regulatory category. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Integrated UAS component stack. **Named evidence/example:** A useful UAS joins airframe, power source, propulsion, flight-control electronics, navigation sensors, communication links, ground or remote pilot station and payload; capability and certification concern the integrated operating system rather than the airframe alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Distinguish UAV, UAS, RPAS and drone, and explain why unmanned does not mean autonomous.…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the sensing-control-actuation loop and its application to stable drone flight. Answer in about 150 words.

Model thesis: Claim: Flight-stabilisation and control chain. **Named evidence/example:** Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Navigation-command-link boundary. **Named evidence/example:** Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Robotics sensing-actuation-control loop. **Named evidence/example:** Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery.
- Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe.
- Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop.

Qualified conclusion: Claim: Flight-stabilisation and control chain. **Named evidence/example:** Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Navigation-command-link boundary. **Named evidence/example:** Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Robotics sensing-actuation-control loop. **Named evidence/example:** Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain the sensing-control-actuation loop and its application to stable drone flight. Answer…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Flight-stabilisation and control chain. **Named evidence/example:** Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Navigation-command-link boundary. **Named evidence/example:** Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Robotics sensing-actuation-control loop. **Named evidence/example:** Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Flight-stabilisation and control chain. **Named evidence/example:** Propulsion creates thrust, control surfaces or rotor-speed changes alter attitude and motion, inertial and other sensors report vehicle state, and the flight controller repeatedly compares sensed state with the commanded path; stable flight therefore depends on a closed control loop, not merely a motor and battery. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Navigation-command-link boundary. **Named evidence/example:** Navigation estimates position, velocity, attitude and route from onboard sensors and available positioning inputs, while the command-and-control link carries pilot commands, telemetry or mission updates; navigation and communication support each other but are not the same subsystem, and loss-of-link behaviour requires a designed failsafe. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Robotics sensing-actuation-control loop. **Named evidence/example:** Robotics connects sensors that perceive state or environment, a controller that interprets inputs and selects commands, actuators that create physical action, and feedback that corrects subsequent action; aerial robotics adds flight dynamics, navigation, communication and airspace constraints to this loop. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain the sensing-control-actuation loop and its application to stable drone flight. Answer…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Compare multirotor, fixed-wing and hybrid VTOL UAVs through mission and payload trade-offs. Answer in about 250 words.

Model thesis: Claim: Payload-mission-capability link. **Named evidence/example:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Multirotor-fixed-wing-hybrid categories. **Named evidence/example:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability.
- Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity.
- Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards.

Qualified conclusion: Claim: Payload-mission-capability link. **Named evidence/example:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Multirotor-fixed-wing-hybrid categories. **Named evidence/example:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **compare** requires a direct position on “Compare multirotor, fixed-wing and hybrid VTOL UAVs through mission and payload trade-offs.…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Payload-mission-capability link. **Named evidence/example:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Multirotor-fixed-wing-hybrid categories. **Named evidence/example:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Payload-mission-capability link. **Named evidence/example:** A payload is the mission device or material carried beyond the flying platform, such as an imaging sensor, mapping instrument, sprayer or delivery load; payload mass, power demand, data rate and integration affect endurance and control, so one platform label cannot establish every mission capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Multirotor-fixed-wing-hybrid categories. **Named evidence/example:** Multirotors use several rotors and suit hover and confined-area work; fixed-wing UAVs generate lift from forward motion and suit wider-area coverage but ordinarily need a launch and recovery solution; hybrid VTOL designs combine vertical take-off or landing with wing-borne cruise at the cost of added integration complexity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Compare multirotor, fixed-wing and hybrid VTOL UAVs through mission and payload trade-offs.…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Examine VLOS, BVLOS, drone-swarm coordination and AI-enabled autonomy as distinct operating challenges. Answer in about 250 words.

Model thesis: Claim: Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AI-enabled autonomy boundary. **Named evidence/example:** Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone-swarm coordination. **Named evidence/example:** A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** VLOS-BVLOS operating boundary. **Named evidence/example:** VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous.
- Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary.
- A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure.
- VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service.

Qualified conclusion: Claim: Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AI-enabled autonomy boundary. **Named evidence/example:** Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone-swarm coordination. **Named evidence/example:** A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** VLOS-BVLOS operating boundary. **Named evidence/example:** VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **examine** requires a direct position on “Examine VLOS, BVLOS, drone-swarm coordination and AI-enabled autonomy as distinct operating…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AI-enabled autonomy boundary. **Named evidence/example:** Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone-swarm coordination. **Named evidence/example:** A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** VLOS-BVLOS operating boundary. **Named evidence/example:** VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Remote operation-automation-autonomy boundary. **Named evidence/example:** Teleoperation keeps a human in direct remote control, automation executes a pre-programmed sequence in a structured setting, and autonomy uses sensing, onboard logic and feedback to select actions toward a goal with reduced direct control; an unmanned aircraft is not automatically autonomous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** AI-enabled autonomy boundary. **Named evidence/example:** Computer vision or other AI can assist perception, classification, route choice and decision support, but an AI output is not itself lawful authority, safe flight or accountable deployment; training data, validation, human oversight, ground truth and defined fallback behaviour remain necessary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone-swarm coordination. **Named evidence/example:** A drone swarm involves multiple aircraft coordinating through communication, localisation and distributed or autonomous control rules; many drones merely flying nearby do not form a swarm, and resilience depends on network design, sensing, task allocation and behaviour under link or member failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** VLOS-BVLOS operating boundary. **Named evidence/example:** VLOS keeps the aircraft within the remote pilot's unaided visual line of sight, whereas BVLOS operates beyond that visual envelope and demands stronger communication, navigation, detect-and-avoid, contingency and airspace-assurance arrangements; an authorised experiment is not general permission for routine BVLOS service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Examine VLOS, BVLOS, drone-swarm coordination and AI-enabled autonomy as distinct operating…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Discuss counter-UAS architecture and the safety, privacy, cybersecurity and liability challenges of dual-use drones. Answer in about 300 words.

Model thesis: Claim: Counter-UAS and electronic-countermeasure layers. **Named evidence/example:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Safety-reliability-liability chain. **Named evidence/example:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Privacy-data-cybersecurity boundary. **Named evidence/example:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation.
- Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone.
- Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience.
- Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment.

Qualified conclusion: Claim: Counter-UAS and electronic-countermeasure layers. **Named evidence/example:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Safety-reliability-liability chain. **Named evidence/example:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Privacy-data-cybersecurity boundary. **Named evidence/example:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **discuss** requires a direct position on “Discuss counter-UAS architecture and the safety, privacy, cybersecurity and liability…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Counter-UAS and electronic-countermeasure layers. **Named evidence/example:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Safety-reliability-liability chain. **Named evidence/example:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Privacy-data-cybersecurity boundary. **Named evidence/example:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Counter-UAS and electronic-countermeasure layers. **Named evidence/example:** Counter-UAS separates detection, identification, tracking, decision and mitigation; radar, radio-frequency, acoustic or optical sensing may contribute, while jamming or spoofing are electronic countermeasures rather than universal solutions, can affect legitimate systems and belong to a security mandate distinct from DGCA civil-safety regulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Safety-reliability-liability chain. **Named evidence/example:** Safe operation links airworthiness, maintenance, pilot or operator competence, weather assessment, route and payload limits, geofencing or other safeguards, command-link resilience, failsafes and incident response; when control is shared among operator, software, manufacturer and service provider, liability must follow evidence about the failed function rather than the word 'autonomous' alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Privacy-data-cybersecurity boundary. **Named evidence/example:** Persistent aerial observation can collect personal, geospatial and operational data, while command, navigation and payload links can face interception, spoofing, jamming or malicious access; registration or a permission portal does not by itself establish purpose limitation, data security, lawful surveillance or cyber resilience. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Discuss counter-UAS architecture and the safety, privacy, cybersecurity and liability…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Critically evaluate India's civil-drone ecosystem through the Drone Rules, Digital Sky, certification, institutions, civilian applications and the approval-to-deployment boundary. Answer in about 300 words.

Model thesis: Claim: Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone Rules category spine. **Named evidence/example:** The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Digital Sky-airspace-institution boundary. **Named evidence/example:** MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Certification-registration-pilot-status ladder. **Named evidence/example:** Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-to-deployment firewall. **Named evidence/example:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards.
- The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval.
- MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system.

- Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions.
- Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment.
- Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status.

Qualified conclusion: Claim: Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone Rules category spine. **Named evidence/example:** The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Digital Sky-airspace-institution boundary. **Named evidence/example:** MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Certification-registration-pilot-status ladder. **Named evidence/example:** Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-to-deployment firewall. **Named evidence/example:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **evaluate** requires a direct position on “Critically evaluate India's civil-drone ecosystem through the Drone Rules, Digital Sky,…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone Rules category spine. **Named evidence/example:** The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Digital Sky-airspace-institution boundary. **Named evidence/example:** MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Certification-registration-pilot-status ladder. **Named evidence/example:** Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-to-deployment firewall. **Named evidence/example:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Civilian application and evidence boundary. **Named evidence/example:** Drones can support crop monitoring or spraying, volcanic observation, wildlife research, surveying, infrastructure inspection, disaster assessment, logistics pilots and public administration by improving access or local observation; imagery still requires permissions, trained interpretation, field verification and sector-specific safeguards. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Drone Rules category spine. **Named evidence/example:** The audited owners preserve maximum all-up-weight categories of nano up to 250 g, micro above 250 g to 2 kg, small above 2 kg to 25 kg, medium above 25 kg to 150 kg and large above 150 kg; category is a regulatory risk proxy, not proof of payload, range, autonomy or mission approval. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Digital Sky-airspace-institution boundary. **Named evidence/example:** MoCA anchors policy and notification, DGCA civil regulation and certification, AAI airspace and air-traffic services, and BCAS aviation security; Digital Sky supports registration, identification, permissions and compliance workflows and publishes the green-yellow-red airspace map, but is not itself an air-traffic-control system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Certification-registration-pilot-status ladder. **Named evidence/example:** Type certification where required evaluates the approved aircraft type, a Unique Identification Number identifies an aircraft, and an RPC through a DGCA-authorised RPTO addresses remote-pilot competence where required; type certificate, registration, pilot credential, airspace permission and lawful mission approval are separate conditions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Dual-use and industrial-policy boundary. **Named evidence/example:** Civil drones, military UAVs, loitering munitions and counter-UAS share enabling technologies but follow distinct civil-regulatory, defence-procurement and security routes; the owners identify drone-and-component PLI and the complete-drone import restriction with stated exceptions as industrial-policy instruments, not proof of indigenous capability or operational deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Approval-to-deployment firewall. **Named evidence/example:** Rule notification, amendment, portal access, type certification, registration, pilot certification, trial authorisation, scheme approval, manufacturing incentive, procurement, delivery, operational deployment and verified outcome are separate evidence rungs; none establishes performance range, sales, approval, field use, safety record or public benefit beyond its own dated status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Critically evaluate India's civil-drone ecosystem through the Drone Rules, Digital Sky,…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.